SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 10%

Duration : 45 Minutes

QUESTIONS: 2 **Total Marks:20**

Annexure A

Descriptive Written Test

Code of Conduct:

I G. Ramani (19232279) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Gosala Ramani

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

CO1_AT3: Descriptive Written Test

CSA1708 - Artificial Intelligence

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1) Smart Home Automation System

A smart home automation system controls lighting, temperature and security using information from the sensors such as motion detectors, temperature sensors, cameras and clocks.

Types of intelligent Agents.

16/20 B

a) Simple reflex agents:

Agent which makes decision based only on the current input without considering past events.

Ex:-

* If motion is detected → Turn on lights.
* If temp. is above 30°C → Turn on air conditioner.
* If no motion is detected for 10 mins → Turn of the lights.

Advantages:-

* Fast response.
* Easy to implement.
* Suitable for simple tasks.

Disadvantages:-

* Cannot remember previous states.
* Cannot learn user preferences.
* May make incorrect decisions in complex situations.

b) Model-Based

Maintains an internal model of the environment and remembers functions states.

Ex :-

→ knows that the user is sleeping at night.

→ keeps track of room Occupancy.

Advantages:-

Improves reliability

Makes better decision using past information.

Disadvantages:-

→ More memory and computation required.

→ More complex than a simple reflex agent.

c) Goal-based agent:-

A Goal-based agent chooses actions that help achieve specific goals.

Ex:- Maintains room temp. at 24°c.

Reduced electricity consumption.

Advantages:-

Makes intelligent decisions based on goals.

Can evaluate different possible actions.

Disadvantages:-

Requires planning

May take more time to make decisions.

d) Utility-Based Agent

A utility based agent selects the action that provides highest overall benefit by considerings multiple factors

Ex:- The system balances

User Comfort
Energy Consumption
Home security
Electricity cost.

Advantages:-

Produces the best overall decision
Handles trade-off effectively.

Disadvantages:-

Most complex to design
Requires utility calculations and learning mechanisms.

Conclusion:-

A hybrid system using model-based and utility based systems agent provides the best balance between comfort, security & energy efficiency.

2) Robotic in an Unknown maze - UCS & IDs.

A robot is placed in an unknown maze & must find the exit without knowing the map. It explores different path until it reaches the goal. Search algorithms help the robot to decide which path to follow.

UCS (Uniform Cost Search)

working:

Always expands the node with the lowest cumulative path cost.

Ex:- If one path costs 5 units & another 4 Costs 8 units, UCS explores the path with 5 units first.

Advantages:-

→ Guarantees the lowest-cost path

→ Complete if path costs are positive

Disadvantages:-

→ High memory usage.

→ Slow for large search spaces.

→ Stores many nodes.

Complexity:-

Time Complexity:- Exponential (depends on branching factor and path cost).

Space Complexity:- Exponential.

Interactive deepening search (IDs)

working:

IDs performs repeated depth-first search with increasing depth limits until the goal is found.

Ex:- Depth 0
Depth 1
Depth 2

* Continue until the exit is discovered.

Advantages:-

→ Uses very little money

→ Finds shallow solutions efficiently.

Disadvantages:-

→ Repeats node expansions.

→ Not suitable when path costs differ.

Complexity:

Time Complexity : $O(b^d)$

Space Complexity : $O(b \times d)$

where b = branching factor

d = depth of the goal.

Conclusion:-

A goal-based agent combined with uniform cost search provides optimal navigation by finding the least-cost path, while interactive deepening search when memory is limited.